Derivation and Implementation Notes

Yi, Huo, Liu, Fan, Zhang & Cao — TTITE Algorithm

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Companion document for the `quarksim` codebase

Contents

1 The Core Idea: Imaginary-Time Evolution **2**
- 1.1 Why Imaginary Time? . . . 2

2 Test Systems **2**
- 2.1 Hydrogen Molecule (Eq. 12) . . . 2
- 2.2 Heisenberg Spin-1/2 Chain (Eq. 13) . . . 2
- 2.3 Initial States . . . 2

3 Trotter Decomposition (Eq. 2) **3**

4 Taylor Expansion and LCU (Eqs. 4–5) **3**
- 4.1 The Key Simplification: $h_i^2 = I$ . . . 3
- 4.2 Matrix-Based Application . . . 3

5 Quantum Circuit Implementation (Algorithm 1) **3**
- 5.1 The 4-Step Circuit . . . 3
- 5.2 Success Probability (Eq. 11) . . . 4

6 Improved Circuit (Section 5, Fig. 7) **4**

7 Results **4**
- 7.1 H_2 Molecule: Convergence (Figure 2) . . . 4
- 7.2 H_2 Molecule: Potential Energy Surface (Figure 3) . . . 5
- 7.3 Heisenberg Chain: Convergence (Figure 4) . . . 5
- 7.4 Heisenberg Chain: Scaling (Figure 5) . . . 5

8 Code–Equation Mapping **5**

9 Running the Simulation **5**

1 The Core Idea: Imaginary-Time Evolution

1.1 Why Imaginary Time?

Given a Hamiltonian H with ground state $|E_0\rangle$ and eigenvalue E_0, the imaginary-time evolution operator $e^{-H\tau}$ preferentially amplifies the lowest-energy component of any initial state:

$$|\psi(\tau)\rangle = A(\tau)\, e^{-H\tau}\, |\psi(0)\rangle \tag{1}$$

where $A(\tau)$ is a normalization constant. Since each eigenstate $|E_k\rangle$ is damped by $e^{-E_k\tau}$, the ground state (smallest E_k) dominates as $\tau \to \infty$.

The imaginary-time Schrödinger equation is (Eq. 1 of the paper):

$$-\frac{d\,|\psi\rangle}{d\tau} = H\,|\psi\rangle \tag{2}$$

The challenge: $e^{-H\tau}$ is **non-unitary**—it cannot be directly implemented as a quantum gate. The TTITE algorithm overcomes this using Trotter decomposition, Taylor expansion, and ancilla-qubit postselection.

2 Test Systems

2.1 Hydrogen Molecule (Eq. 12)

The H_2 molecular Hamiltonian in the STO-3G basis, after Jordan-Wigner transformation and qubit reduction to 2 qubits:

$$H_{\mathrm{H}_2} = c_0(D)\, I \otimes I + c_1(D)\, Z \otimes I + c_2(D)\, I \otimes Z + c_3(D)\, Z \otimes Z + c_4(D)\, X \otimes X \tag{3}$$

where the coefficients $c_i(D)$ depend on the interatomic distance D.

Parameters (Table 1): The coefficients are computed from PySCF (RHF + molecular orbital integrals), mapped to a 2-qubit Hamiltonian via the FCI singlet/triplet block structure.

D (Å)	c_0	$c_1 = c_2$	c_4
0.35	0.701	−0.747	0.163
0.735	−0.332	−0.398	0.181

Table 1: Selected H_2 Hamiltonian coefficients (c_3 is small, ~ 0.01).

Code: `hamiltonian.h2_coefficients(D)` interpolates from a 47-entry table computed via PySCF.

2.2 Heisenberg Spin-1/2 Chain (Eq. 13)

$$H_{\mathrm{Heisenberg}} = -J \sum_{j=1}^{n-1} \left(\sigma_x^j \sigma_x^{j+1} + \sigma_y^j \sigma_y^{j+1} + \sigma_z^j \sigma_z^{j+1}\right) - h \sum_{j=1}^{n} \sigma_z^j \tag{4}$$

where J is the coupling strength, h is the static magnetic field, and open boundary conditions are used.

Code: `hamiltonian.build_heisenberg_hamiltonian(n, J, h)`.

2.3 Initial States

Both systems use $|+\rangle^{\otimes n}$ as the initial state, which has a finite overlap with the ground state. For Figure 3, a biased initial state $(4\,|00\rangle + |01\rangle + |10\rangle + |11\rangle)/\sqrt{19}$ is used.

3 Trotter Decomposition (Eq. 2)

For $H = \sum_{i=1}^{m} c_i h_i$ (sum of Pauli product terms), the Trotter approximation decomposes:

$$e^{-H\tau} = \left(e^{-c_1 h_1 \Delta\tau} \cdots e^{-c_m h_m \Delta\tau}\right)^L + O(\Delta\tau) \tag{5}$$

where $\Delta\tau$ is the Trotter step duration and $L = \tau/\Delta\tau$ is the number of segments.

Each factor $S(\Delta\tau) = \prod_i e^{-c_i h_i \Delta\tau}$ is called a **Trotter segment**, and each $T_i(\Delta\tau) = e^{-c_i h_i \Delta\tau}$ is a **Trotter step**.

The Trotter error is $O(\Delta\tau)$ per segment. As shown in the paper's Figure 6, this is the dominant source of error—the Taylor expansion error is negligible in comparison.

Code: `evolution.apply_trotter_segment_matrix()` iterates over all Hamiltonian terms.

4 Taylor Expansion and LCU (Eqs. 4–5)

4.1 The Key Simplification: $h_i^2 = I$

Since each h_i is a Pauli product (tensor product of $\{I, X, Y, Z\}$), we have $h_i^2 = I$. This means the Taylor series of $e^{-c_i h_i \Delta\tau}$ collapses:

$$T_i^R(\Delta\tau) = \sum_{j=0}^{R} \frac{(-c_i\Delta\tau)^j}{j!} h_i^j = \underbrace{\sum_{\substack{j=0 \\ j \text{ even}}}^{R} \frac{(-c_i\Delta\tau)^j}{j!}}_{\alpha} I + \underbrace{\sum_{\substack{k=0 \\ k \text{ odd}}}^{R} \frac{(-c_i\Delta\tau)^k}{k!}}_{\beta} h_i \tag{6}$$

So $T_i(\Delta\tau) \approx \alpha I + \beta h_i$ — a linear combination of just **two** unitaries (I and h_i).

When $R \to \infty$: $\alpha \to \cosh(c_i\Delta\tau)$ and $\beta \to -\sinh(c_i\Delta\tau)$.

Code: `circuit.taylor_coefficients(c_i, dt, order)` computes (α, β).

4.2 Matrix-Based Application

In our statevector simulation, each Trotter step simply applies:

$$\left|\psi'\right\rangle = (\alpha I + \beta h_i)\left|\psi\right\rangle, \qquad \left|\psi_{\text{norm}}\right\rangle = \frac{\left|\psi'\right\rangle}{\| \left|\psi'\right\rangle \|} \tag{7}$$

Since $\alpha I + \beta h_i$ preferentially dampens high-energy eigenstates of h_i (relative to the sign of c_i), the product of all Trotter steps across all terms drives the state toward the ground state of H.

Code: `evolution.apply_trotter_step_matrix()`.

5 Quantum Circuit Implementation (Algorithm 1)

On a quantum computer, the non-unitary operator $\alpha I + \beta h_i$ is implemented using one **ancilla qubit** and the Linear Combination of Unitaries (LCU) technique.

5.1 The 4-Step Circuit

1. **Encoding:** Apply $R_y(\theta)$ on the ancilla, where $\theta = 2\arccos\left(\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}\right)$:

$$\left|\phi_1\right\rangle = \left(\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}\left|0\right\rangle + \frac{\beta}{\sqrt{\alpha^2+\beta^2}}\left|1\right\rangle\right)\left|\psi\right\rangle \tag{8}$$

2. **Controlled-h_i:** Apply $CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes h_i$:

$$|\phi_2\rangle = \frac{1}{\sqrt{\alpha^2+\beta^2}} \left(\alpha\,|0\rangle\,|\psi\rangle + \beta\,|1\rangle\,h_i\,|\psi\rangle\right) \tag{9}$$

   Since h_i is a Pauli product, CU decomposes into controlled-X, controlled-Y, controlled-Z gates.
3. **Hadamard:** Apply H on the ancilla:

$$|\phi_3\rangle = \frac{1}{\sqrt{2(\alpha^2+\beta^2)}} \left[|0\rangle\,(\alpha I + \beta h_i)\,|\psi\rangle + |1\rangle\,(\alpha I - \beta h_i)\,|\psi\rangle\right] \tag{10}$$

4. **Measure:** If ancilla $= |0\rangle$, the work system collapses to $(\alpha I + \beta h_i)\,|\psi\rangle$ (normalized).

5.2 Success Probability (Eq. 11)

The probability of obtaining $|0\rangle$ on the ancilla is:

$$P_s = \frac{1}{2(\alpha^2+\beta^2)} \|(\alpha I + \beta h_i)\,|\psi\rangle\,\|^2 \tag{11}$$

Lemma 1: The success probability is monotonically increasing—$P(\Delta\tau) \le P(2\Delta\tau) \le \cdots \le P(L\Delta\tau)$.

Code: `circuit.build_trotter_step_circuit()` builds the full LCU circuit for one step.

6 Improved Circuit (Section 5, Fig. 7)

The standard circuit's success probability can be improved by replacing the Hadamard in Step 3 with $R_y(\theta')^\dagger$, where:

$$\theta' = 2\arccos\left(\sqrt{\frac{\alpha}{\alpha+\beta}}\right) \tag{12}$$

The improved success probability is:

$$P_s' = \frac{1}{(\alpha+\beta)^2} \|T_i(\Delta\tau)\,|\psi\rangle\,\|^2 \ge P_s \tag{13}$$

since $(\alpha+\beta)^2 \le 2(\alpha^2+\beta^2)$.

Code: `circuit.build_improved_trotter_step_circuit()`.

7 Results

7.1 H_2 Molecule: Convergence (Figure 2)

Quantity	Our code	Paper
E_0 (exact)	-0.7893	~ -0.85
E_τ (TTITE, $\tau = 3$)	-0.7868	$\approx E_0$
Fidelity ($\tau = 3$)	0.999	≈ 1.0

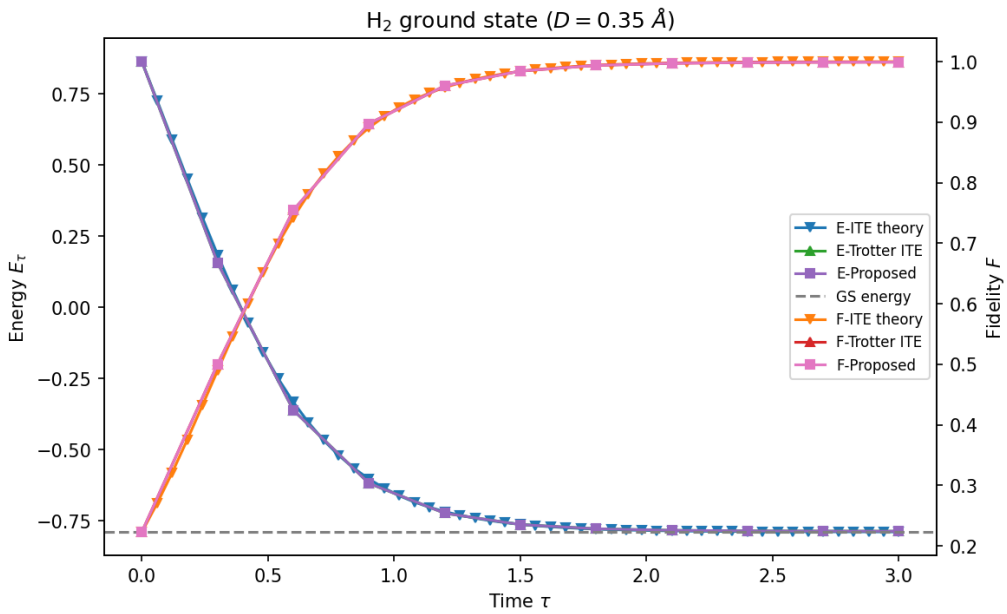

Figure 1: Energy E_τ and fidelity F vs. imaginary time τ for H_2 at $D = 0.35$ Å. Three methods are compared: exact ITE (theory), Trotter ITE (no Taylor error), and the proposed TTITE (5th-order Taylor). All converge to the ground state by $\tau \approx 2$.

7.2 H_2 Molecule: Potential Energy Surface (Figure 3)

7.3 Heisenberg Chain: Convergence (Figure 4)

7.4 Heisenberg Chain: Scaling (Figure 5)

8 Code–Equation Mapping

Paper equation	Code location	Description
Eq. (1): $-d\,\lvert\psi\rangle / d\tau = H\,\lvert\psi\rangle$	`evolution.run_ite_exact()`	Exact ITE
Eq. (2): Trotter decomposition	`evolution.run_trotter_ite()`	Trotter ITE
Eqs. (4–5): Taylor expansion	`circuit.taylor_coefficients()`	α, β coefficients
Eq. (7): Controlled unitary	`circuit.build_trotter_step_circuit()`	LCU circuit
Eq. (11): Success probability	`evolution.apply_trotter_step_matrix()`	P_s computation
Eq. (12): H_2 Hamiltonian	`hamiltonian.build_h2_hamiltonian()`	2-qubit H_2
Eq. (13): Heisenberg chain	`hamiltonian.build_heisenberg_hamiltonian()`	n-qubit chain
Eqs. (18–20): Improved circuit	`circuit.build_improved_trotter_step_circuit()`	Higher P_s'

9 Running the Simulation

```
# Install
uv venv && source .venv/bin/activate
uv pip install -e .

# Reproduce all paper figures
python -m quarksim.yihuoliufanzhang.run

# Single figure
python -m quarksim.yihuoliufanzhang.run --figure 2
```

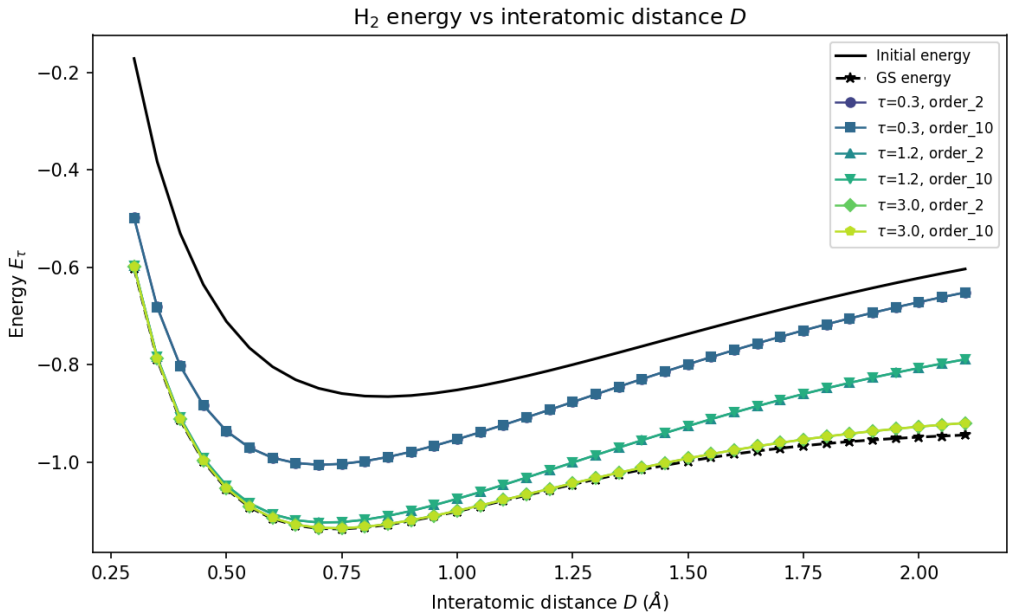

Figure 2: H_2 energy after TTITE as a function of interatomic distance D, for different (τ, R) combinations. As τ increases and/or the Taylor order R increases, the TTITE energy approaches the exact ground state energy (dashed). The PES minimum occurs at $D \approx 0.70$–0.75 Å, consistent with the known equilibrium bond length.

```
# Heisenberg chain
python -m quarksim.yihuoliufanzhang.run --system heisenberg --n 6

# Custom parameters
python -m quarksim.yihuoliufanzhang.run --system h2 --D 0.735 --tau 5
   --dt 0.1 --order 10
```

Output files:

- `ttite_result.json` — full simulation record
- `figure2_h2_convergence.png` — H_2 energy/fidelity convergence
- `figure3_h2_pes.png` — H_2 potential energy surface
- `figure4_heisenberg_convergence.png` — Heisenberg convergence
- `figure5_heisenberg_scaling.png` — Heisenberg multi-size scaling

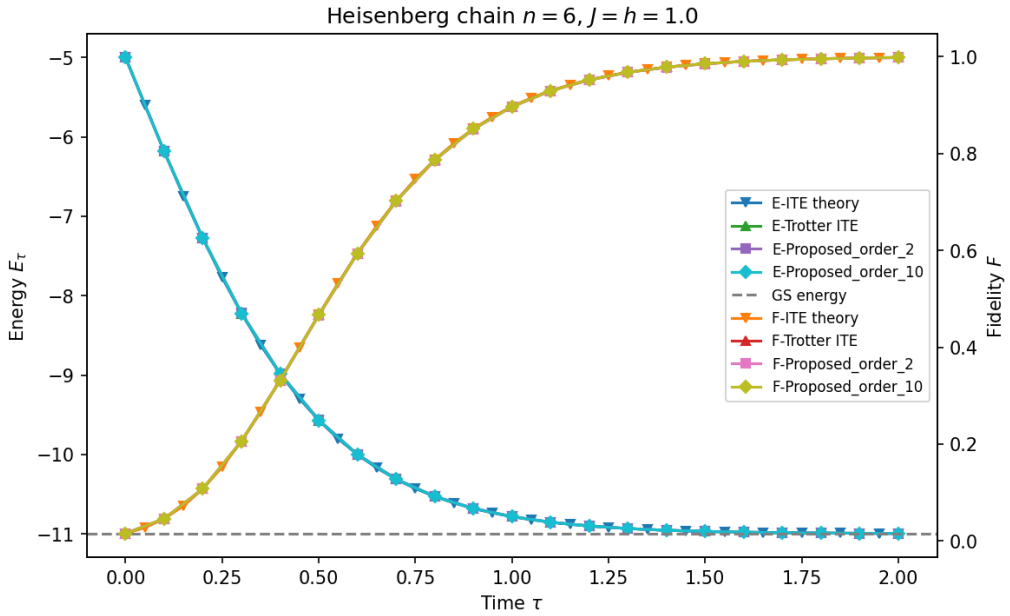

Figure 3: Heisenberg chain with $n = 6$, $J = h = 1$. Energy and fidelity converge rapidly. The 2nd and 10th-order Taylor expansions give nearly identical results, confirming that the Taylor error is negligible compared to the Trotter error.

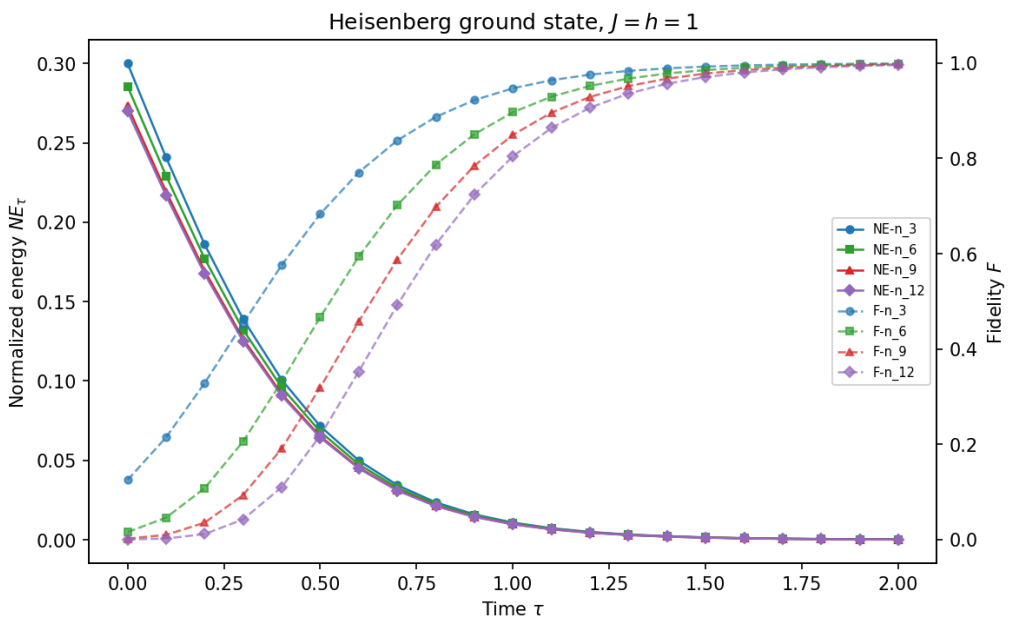

Figure 4: Normalized energy $NE_\tau = (E_\tau - E_{\min})/(E_{\max} - E_{\min})$ and fidelity F for Heisenberg chains of different sizes ($n = 3, 6, 9, 12$). The algorithm successfully finds the ground state at all sizes, demonstrating scalability.